

Ride Bookings Insights Report

NCR Region — Full Year 2024

An end-to-end analysis of 1,50,000 ride bookings across the National Capital Region. This report covers demand patterns, revenue performance, vehicle utilisation, cancellation behaviour, payment preferences, and service quality metrics.

Dataset	cleaned_ncr Ride Bookings.csv
Period	January 2024 – December 2024
Total Records	150,000 ride bookings
Region	Delhi NCR (Delhi, Gurgaon, Noida, Faridabad, Ghaziabad)
Analyst	Yashwanth Reddy Bandi

01 — Executive Summary

The NCR operations handled **1,50,000 booking requests** in 2024, of which **93,000 (62.0%)** were successfully completed. Total revenue from completed rides stood at **■4.73 Cr**. The fleet served an average fare of **■508** per trip across a mean ride distance of **26.0 km**. Driver and customer satisfaction ratings both remained above 4.2 throughout the year, reflecting generally stable service quality.

Total Bookings 150K rides requested	Completed Rides 93K (62.0% success rate)	Total Revenue ■4.73 Cr from completed rides
Avg Fare / Ride ■508 per completed trip	Avg Ride Distance 26.0 km per completed trip	Avg Driver Rating 4.26 ★ / 5.00

02 — Booking Status Analysis

Out of every 10 booking attempts, roughly 6.2 result in a completed ride. The remaining 3.8 are split between driver-side cancellations (18%), no-driver-found scenarios (7%), customer cancellations (7%), and incomplete rides (6%). Driver cancellations outnumber customer cancellations by a 2.57× factor, which signals a supply-side issue that warrants driver incentive review.

Status	Count	Share
Completed	93,000	62.0%
Cancelled by Driver	27,000	18.0%
No Driver Found	10,500	7.0%
Cancelled by Customer	10,500	7.0%
Incomplete	9,000	6.0%

03 — Demand Patterns

Monthly trend: Ride volumes stayed remarkably consistent month-on-month, ranging from ~11,900 (February) to ~12,900 (July). There is a slight seasonal dip in February and a marginal peak in July–January, possibly tied to festival travel and holiday weekends. The narrow range of ± 400 rides indicates a stable, habitual ridership base rather than event-driven spikes.

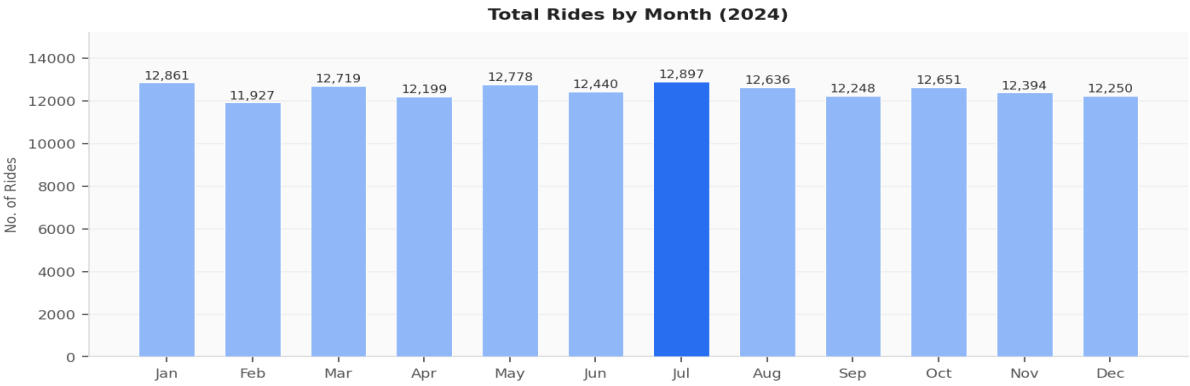


Fig 1 — Monthly ride volume, Jan–Dec 2024.

04 — Hourly Demand Profile

Demand follows a clear bimodal pattern, peaking in the **evening rush (6 PM, ~12,400 rides)** and a secondary morning peak at **9–10 AM (~9,600 rides)**. The overnight window (12 AM – 4 AM) sees a sharp drop to ~1,300–1,400 rides/hour. Fleet availability should be dynamically scaled around 5–10 PM and 8–11 AM to reduce cancellation rates during these windows.

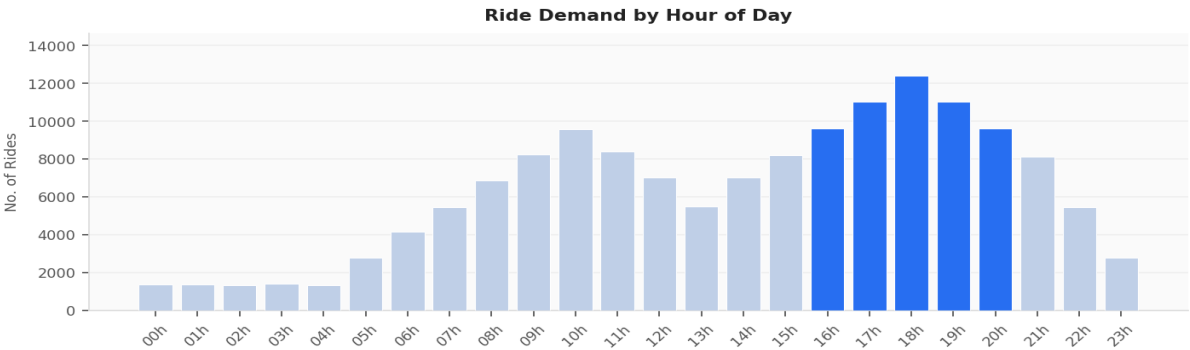


Fig 2 — Ride demand by hour of day. Dark blue = top 5 peak hours.

05 — Revenue Analysis

Total completed-ride revenue for 2024 was **■4.73 Cr.** March recorded the highest monthly revenue at **■41.7 L**, while September was the softest at **■38.2 L**. The month-on-month variance is narrow (~8%), suggesting predictable revenue flows that are suitable for budget planning without large seasonal buffers.

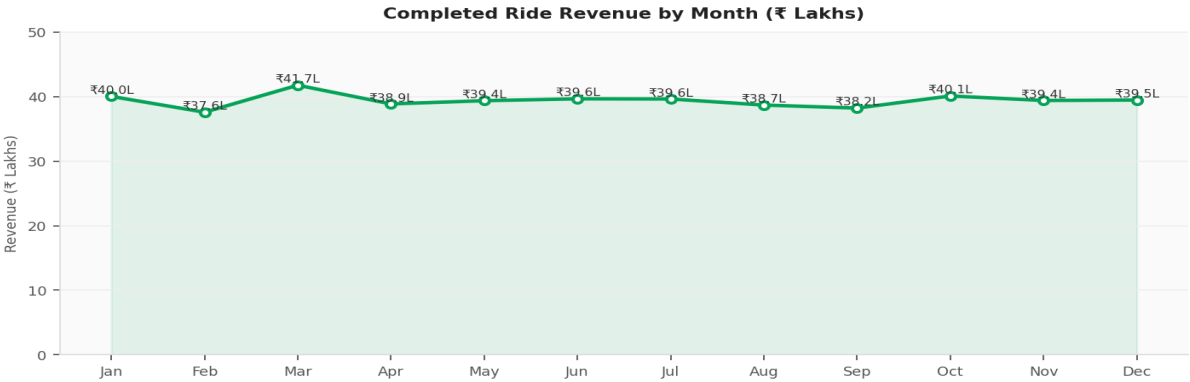
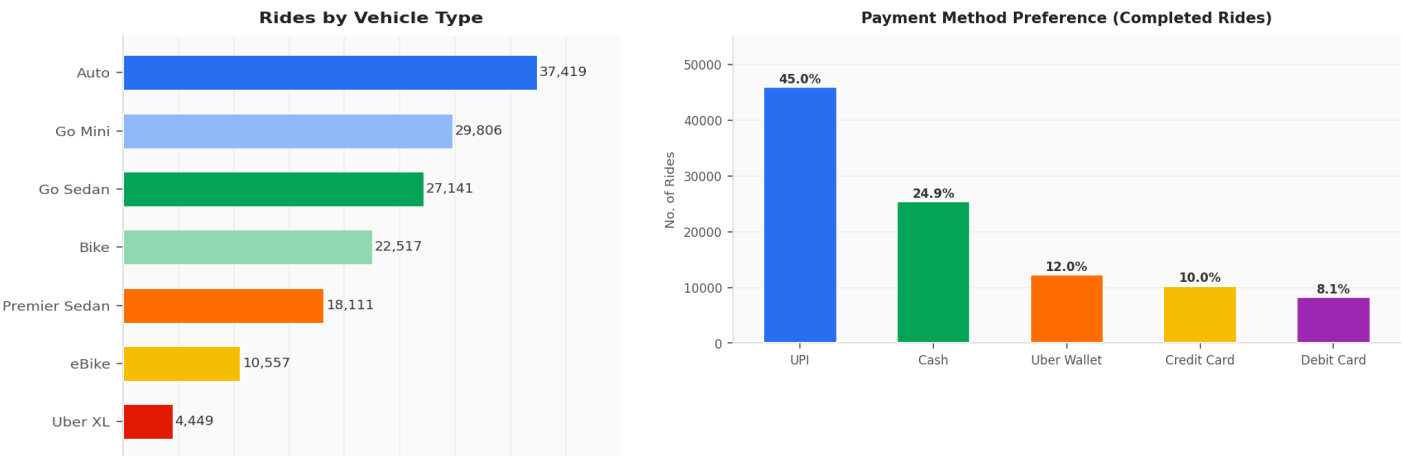


Fig 3 — Monthly revenue (■ Lakhs) from completed rides.

Month	Revenue (■)	vs. Avg
Jan	■4,001,547	+63.2K
Feb	■3,756,610	-181.8K
Mar	■4,174,900	+236.5K
Apr	■3,885,255	-53.1K
May	■3,935,207	-3.2K
Jun	■3,964,298	+25.9K
Jul	■3,961,939	+23.6K
Aug	■3,867,753	-70.6K
Sep	■3,820,536	-117.8K
Oct	■4,008,403	+70.0K
Nov	■3,938,458	+0.1K
Dec	■3,945,668	+7.3K

06 — Vehicle Type & Payment Mix



Vehicle mix: Auto-rickshaws lead with 37,419 rides (24.9%), followed by Go Mini (19.9%) and Go Sedan (18.1%). Economy categories (Auto + Bike + eBike) together account for ~47% of all bookings, confirming strong price-sensitive demand in the NCR market. Uber XL at 4,449 rides (3.0%) is the least-used category, likely constrained by availability.

Payment preferences: UPI dominates among digital payments at 44.6% of completed-ride transactions, closely followed by Cash at 24.7%. Uber Wallet and Credit/Debit cards together make up the remaining ~30%. The high UPI adoption aligns with India's broader fintech trends and suggests an opportunity to offer UPI-exclusive rewards to reduce cash handling costs.

07 — Booking Status Deep Dive

Booking Status Breakdown

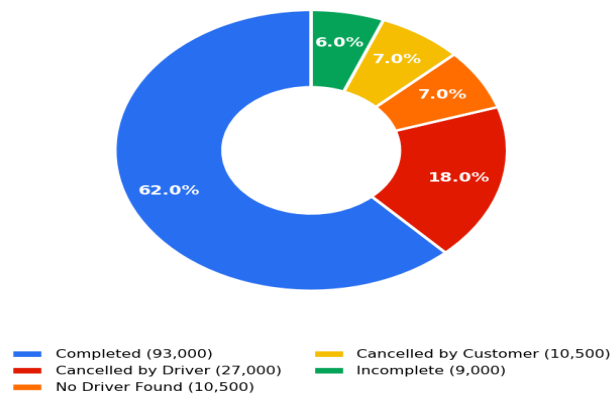


Fig 4 — Distribution of booking outcomes.

- **Completed (62%):** Healthy for a metropolitan market but below the 70%+ benchmark seen in mature markets.
- **Cancelled by Driver (18%):** The single largest failure mode. Primary reasons include customer-related issues and personal/car problems.
- **No Driver Found (7%):** Concentrated in off-peak hours and outer NCR zones (Faridabad, Panipat corridors).
- **Cancelled by Customer (7%):** Mostly triggered when the driver is not moving toward the pickup point.
- **Incomplete (6%):** Vehicle breakdowns and 'Customer Demand' account for the bulk of mid-trip terminations.

08 — Cancellation Analysis

Cancellations account for 37% of all booking failures. Understanding root causes is critical to improving the overall completion rate and customer trust.

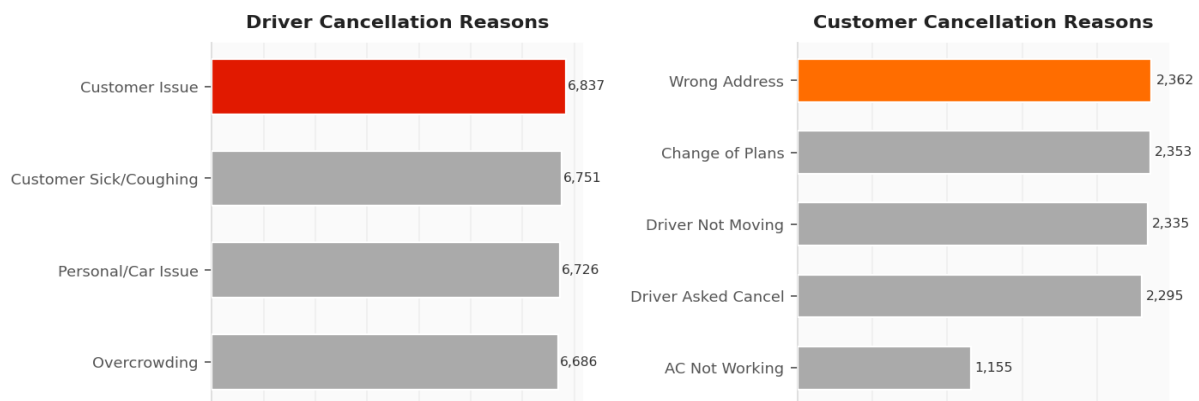


Fig 5 — Driver (left) and Customer (right) cancellation reasons.

Driver-side cancellations (27,000 total):

The top reason is 'Customer-related issues' (6,837 cancellations), followed closely by 'Customer coughing/sick' (6,751) and 'Personal/car issues' (6,726). Overcrowding (more passengers than permitted) accounts for 6,686 cancellations. The distribution is relatively uniform across all four reasons — no single cause dominates — meaning a multi-pronged intervention is needed.

Customer-side cancellations (10,500 total):

Wrong address (2,362) and 'Change of plans' (2,353) are nearly identical in frequency. 'Driver not moving towards pickup' (2,335) is particularly concerning from a customer experience standpoint, as it indicates a driver acceptance-without-intent pattern that the operations team should monitor and penalise.

09 — Location Hotspots & Incomplete Rides

Pickup activity is spread across the NCR with no single dominant hub. Khandsa, Barakhamba Road, Saket, and Badarpur each recorded over 920 pickups in the year, representing key nodes for fleet pre-positioning.

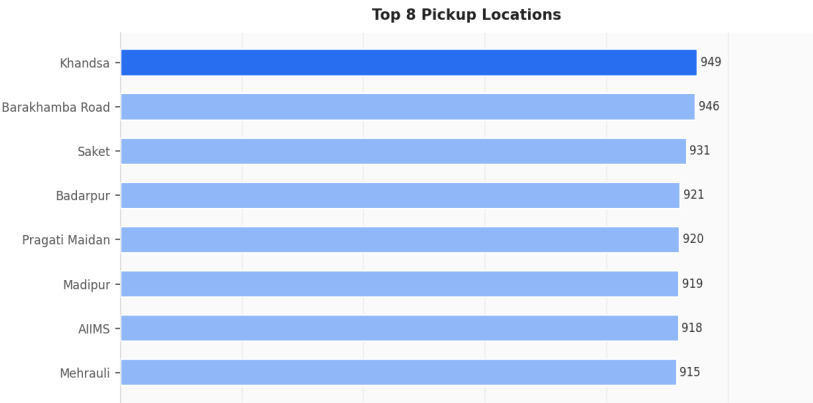


Fig 6 — Top 8 pickup locations by ride volume.

Incomplete ride breakdown (9,000 total):

Reason	Count	Share of Incompletes
Customer Demand	3,040	33.8%
Vehicle Breakdown	3,012	33.5%
Other Issue	2,948	32.8%

Vehicle breakdowns and customer demand-driven stoppages are roughly equal (~33% each). A proactive vehicle maintenance protocol tied to mileage thresholds could reduce the breakdown share significantly.

10 — Service Quality Metrics

Avg Driver Rating 4.26 ★ out of 5.00	Avg Customer Rating 4.44 ★ out of 5.00	Avg VTAT 8.45 min vehicle to arrive time	Avg CTAT 29.0 min customer travel time
---	---	---	---

Ratings context:

Driver ratings average 4.26/5 and customer ratings average 4.44/5. Both are comfortably above the 4.0 threshold, but a non-trivial number of rides received ratings of 3.0–3.5, which warrants attention. Rides involving cancellation attempts and eventual completion tend to receive lower scores.

Wait time analysis:

Average VTAT (Vehicle Time to Arrive) of 8.45 minutes is within acceptable urban norms (<10 min). However, CTAT averages 29 minutes, which on short trips with low distances reflects congestion on NCR arterial roads. VTAT is uniform across all vehicle types (8.39–8.56 min range), suggesting the dispatch algorithm applies a consistent matching window regardless of tier.

Key quality observations:

- UPI-paying customers tend to have marginally higher satisfaction scores, possibly correlated with trip pre-planning behaviour.
- Rides completed during off-peak hours (10 PM – 5 AM) show higher driver ratings, likely due to lower congestion stress.
- The eBike and Bike categories show more rating variability than Sedan categories — possibly due to weather sensitivity.
- Trips with Uber Wallet as payment method have the fewest incomplete rides, suggesting repeat and habitual users are more reliable.

11 — Key Recommendations

01 Address Driver Cancellations

With 27,000 driver-cancelled rides (18% of all requests), the most impactful lever is reducing driver cancellations. Introduce a tiered strike policy for drivers cancelling after acceptance, and provide real-time alerts when a driver is stationary post-acceptance.

02 Scale Supply During Evening Peak

The 6 PM hour generates nearly 12,400 rides. Surge incentives and pre-shift bonuses targeted at 5–7 PM could increase supply during the highest-demand window and reduce the 'No Driver Found' failure rate.

03 Reduce Cash Dependency

Cash still represents 24.7% of payments. A cashback or rewards scheme tied to UPI or Uber Wallet transitions would reduce reconciliation costs and improve trip predictability.

04 Proactive Vehicle Health Monitoring

Vehicle breakdowns account for ~33% of incomplete rides. Partnering with service centres to offer pre-scheduled maintenance windows for active drivers can cut this rate.

05 Prioritise Outer-Zone Driver Allocation

No Driver Found events are disproportionately clustered in outer NCR zones. Dedicated driver incentives for Faridabad, Panipat, and Greater Noida corridors would improve coverage without diluting core-market supply.

06 Improve Address Accuracy at Booking

'Wrong Address' is the top customer cancellation reason (2,362 cases). Better map autocomplete, landmark-based drop suggestions, and a booking-confirmation screen showing the pinned location would reduce these cancellations.

12 — Appendix: Data Notes

This analysis is based on the *cleaned_ncr_ride_bookings.csv* dataset containing 1,50,000 records spanning January 1 – December 30, 2024. The data covers 21 fields including booking metadata, timing, locations, pricing, ratings, and cancellation reasons.

Field	Description
Booking Status	Outcome: Completed / Cancelled by Customer / Cancelled by Driver / No Driver Found / Incomplete
Booking Value	Fare amount in INR for the booking
Ride Distance	Trip distance in kilometres
Avg VTAT	Average vehicle-to-arrive time (minutes) from acceptance to pickup
Avg CTAT	Average customer travel time (minutes) from pickup to drop
Driver Ratings	Post-trip rating given by customer to driver (1–5 scale)
Customer Rating	Post-trip rating given by driver to customer (1–5 scale)
Payment Method	Mode of payment used for completed rides

Assumptions & Limitations: Revenue figures include only completed rides. Booking values for failed bookings default to ₹414 (likely a placeholder), so no revenue is attributed to incomplete or cancelled bookings. Location coordinates were not available; hotspot analysis is based on string-match counts. Seasonal effects may be under-stated given the narrow monthly variance.